

Fault Detection Electrical Models MLflow Report

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1 Overall Model Comparison

This section compares the overall performance of all evaluated models using standard classification metrics.

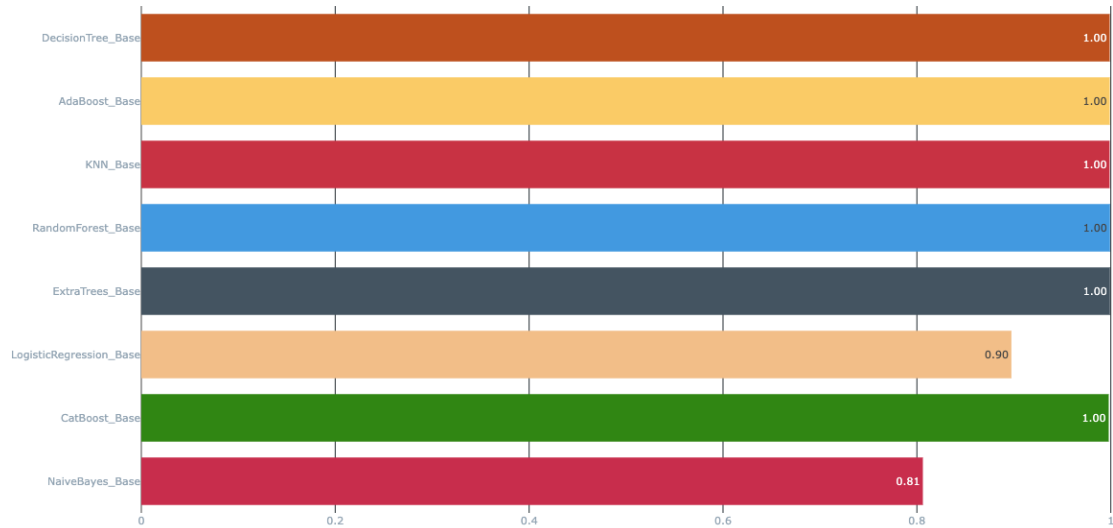


Figure 1: Comparison of Accuracy Across Models

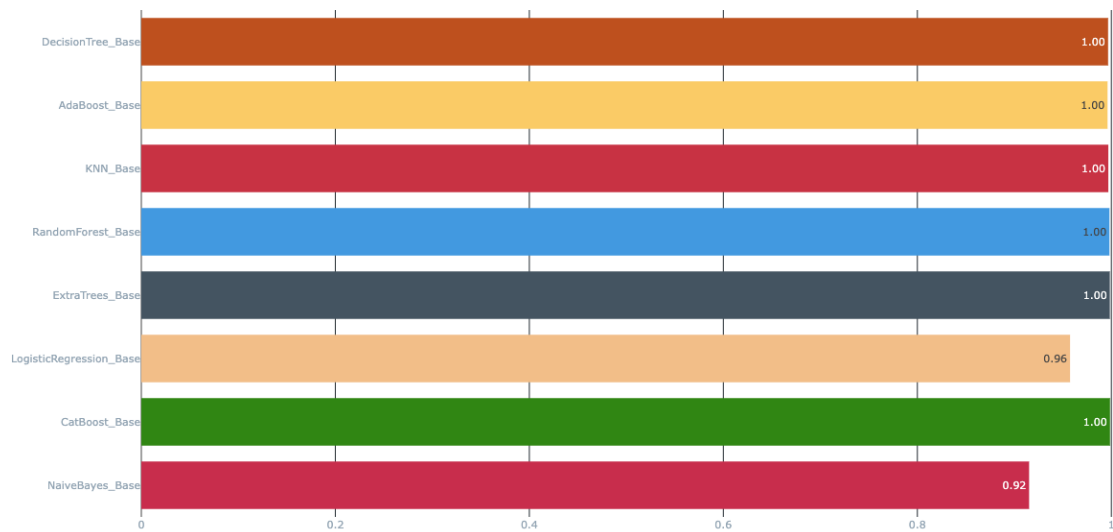


Figure 2: Comparison of Balanced Accuracy Across Models

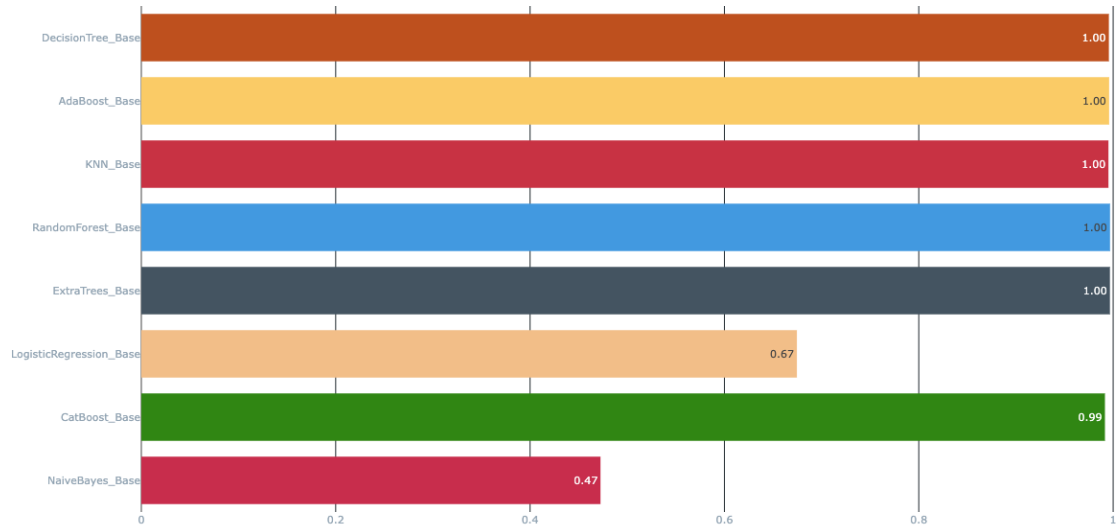


Figure 3: Comparison of Cohen's Kappa Across Models

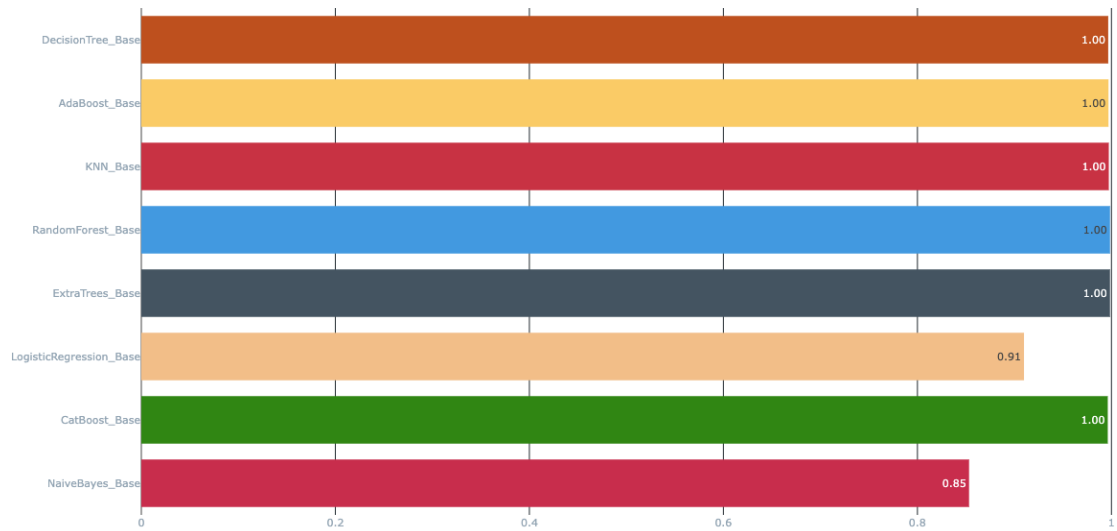


Figure 4: Comparison of F1-Score (Macro) Across Models

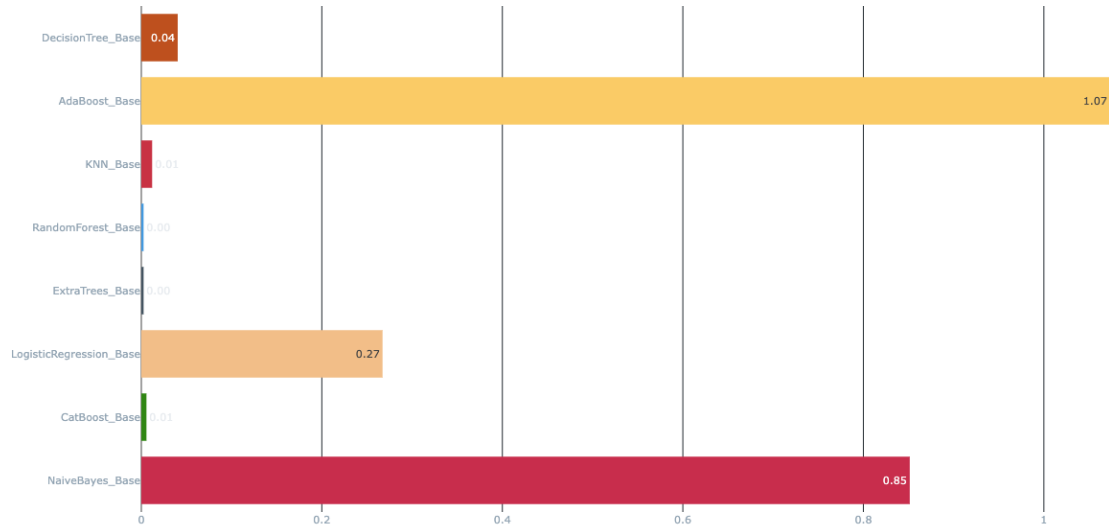


Figure 5: Comparison of Log Loss Across Models

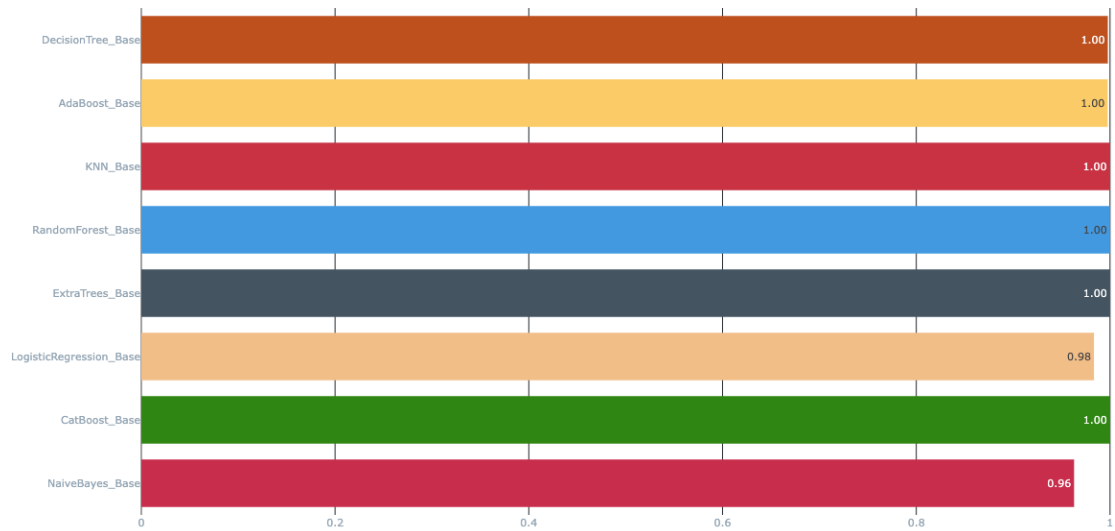


Figure 6: Comparison of ROC-AUC (Macro) Across Models

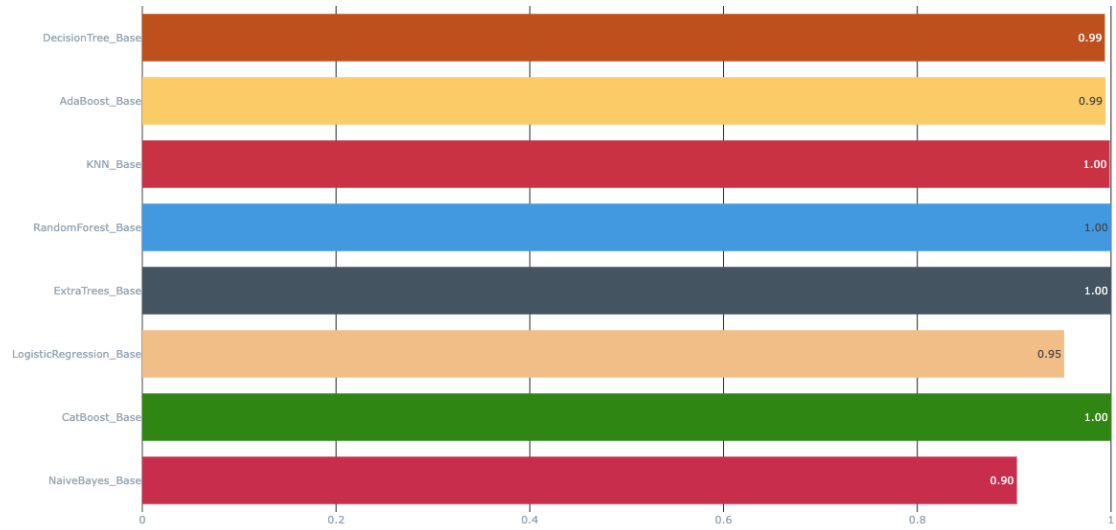


Figure 7: Comparison of PR-AUC (Macro) Across Models

2 Detailed Model Results

This section provides detailed evaluation metrics and plots for each model.

2.1 AdaBoost

Table 1: AdaBoost: Evaluation Metrics

Metric	Value
accuracy	0.996
balanced accuracy	0.998
precision macro	0.996
recall macro	0.997
f1 macro	0.996
cohen kappa	1.073
log loss	0.998
roc auc macro	0.994
pr auc macro	0.994

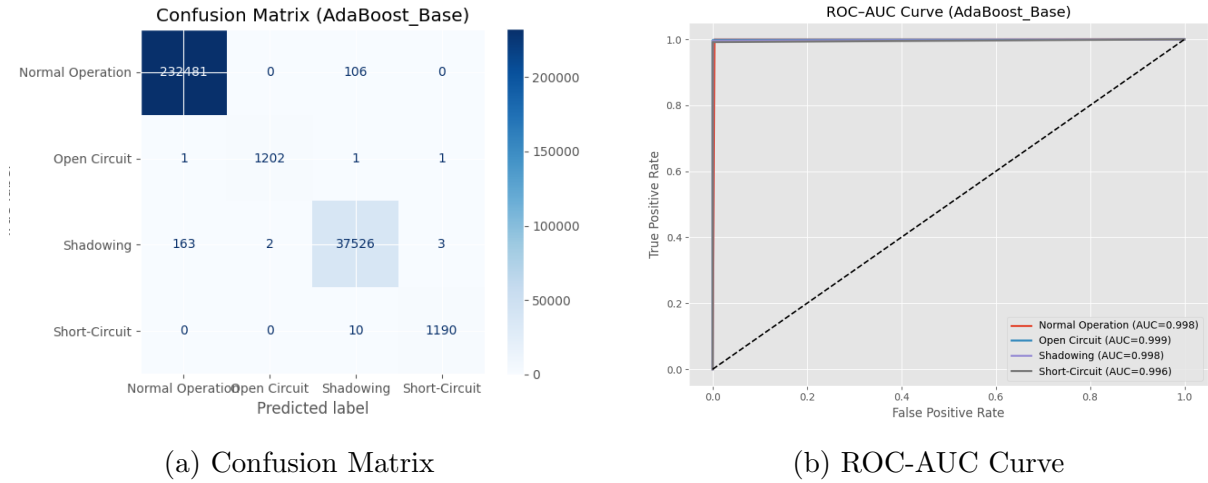


Figure 8: AdaBoost: Classification Performance

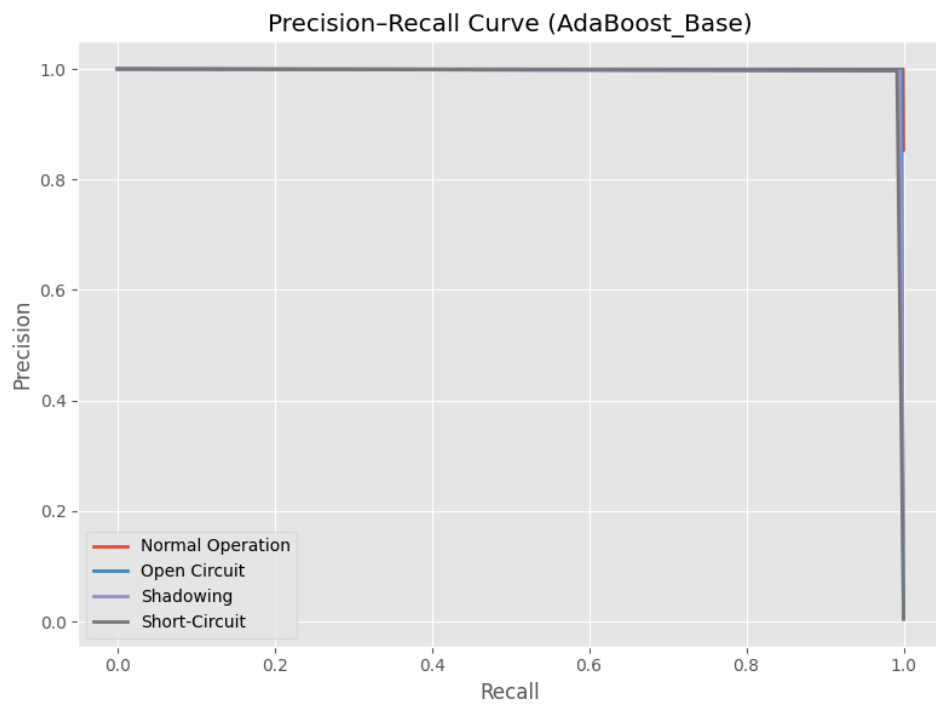


Figure 9: AdaBoost: Precision–Recall Curve

2.2 CatBoost

Table 2: CatBoost: Evaluation Metrics

Metric	Value
accuracy	0.999
balanced accuracy	0.994
precision macro	0.999
recall macro	0.996
f1 macro	0.992
cohen kappa	0.006
log loss	1.000
roc auc macro	1.000
pr auc macro	1.000

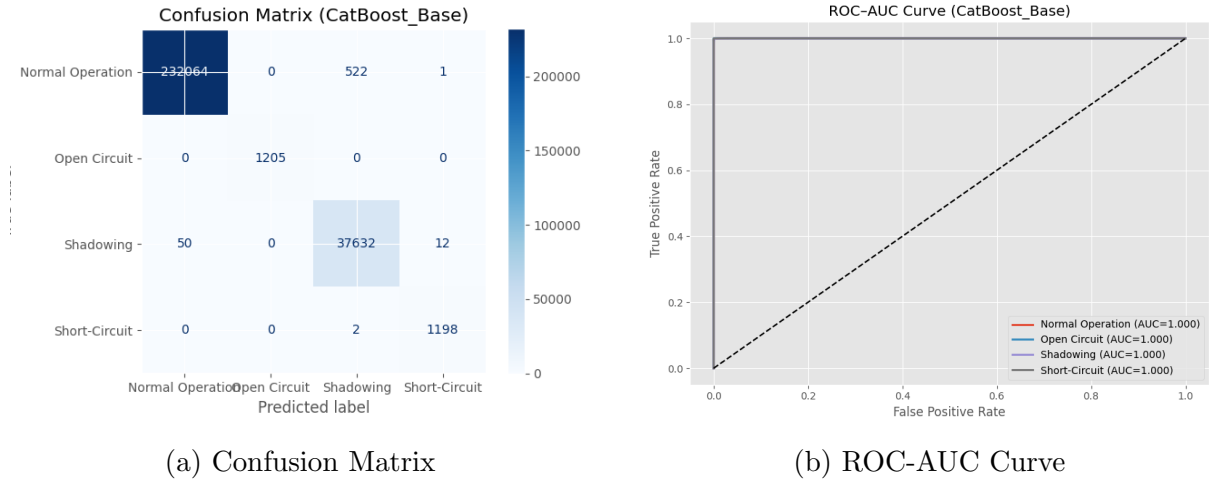


Figure 10: CatBoost: Classification Performance

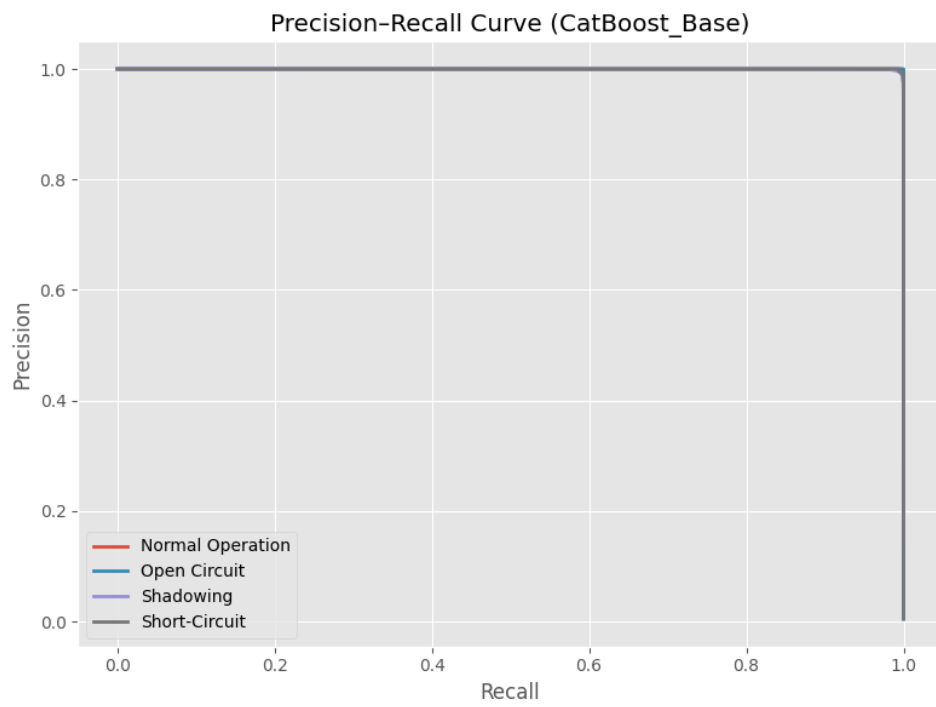


Figure 11: CatBoost: Precision–Recall Curve

2.3 K-Nearest Neighbours

Table 3: K-Nearest Neighbours: Evaluation Metrics

Metric	Value
accuracy	0.997
balanced accuracy	0.998
precision macro	0.997
recall macro	0.997
f1 macro	0.995
cohen kappa	0.012
log loss	1.000
roc auc macro	0.999
pr auc macro	0.999

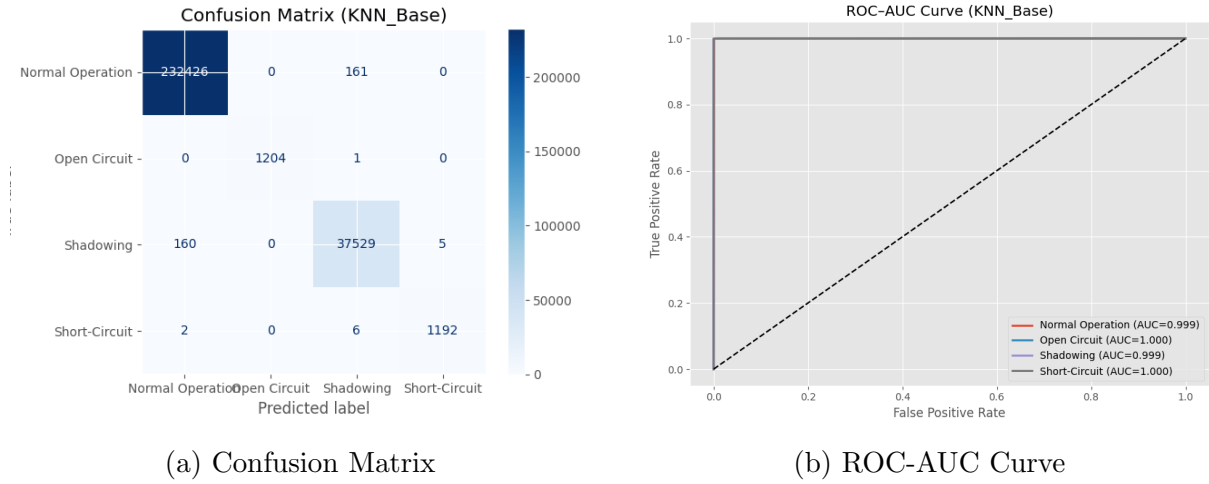


Figure 12: K-Nearest Neighbours: Classification Performance

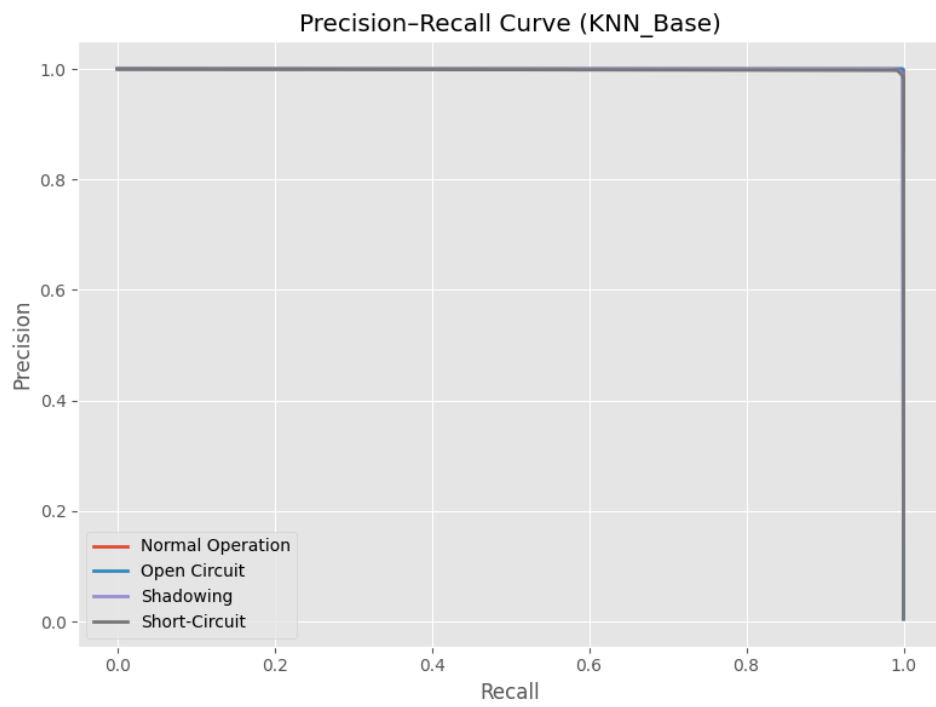


Figure 13: K-Nearest Neighbours: Precision-Recall Curve

2.4 Naive Bayes

Table 4: Naive Bayes: Evaluation Metrics

Metric	Value
accuracy	0.915
balanced accuracy	0.839
precision macro	0.915
recall macro	0.853
f1 macro	0.472
cohen kappa	0.851
log loss	0.963
roc auc macro	0.903
pr auc macro	0.903

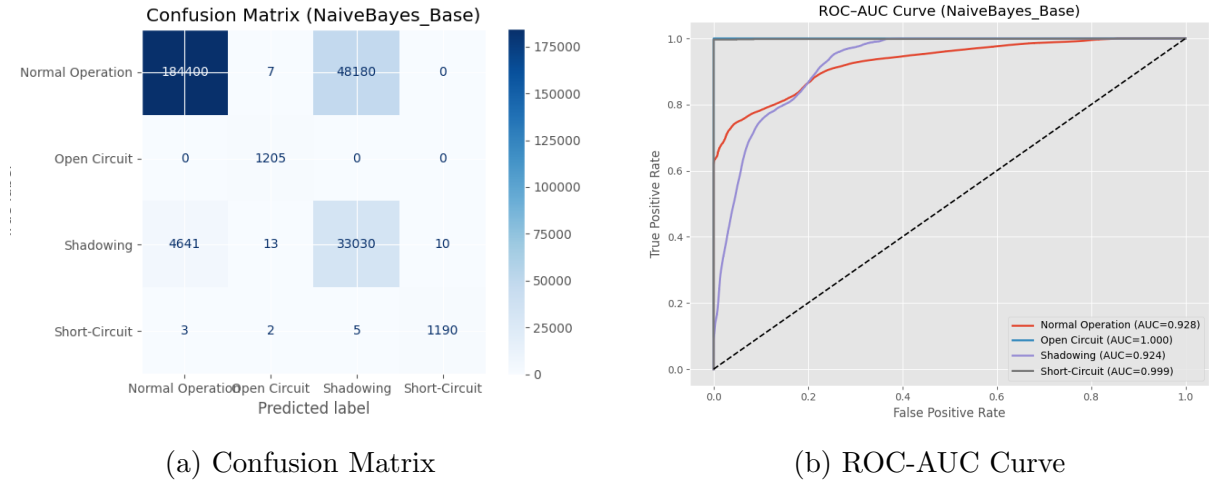


Figure 14: Naive Bayes: Classification Performance

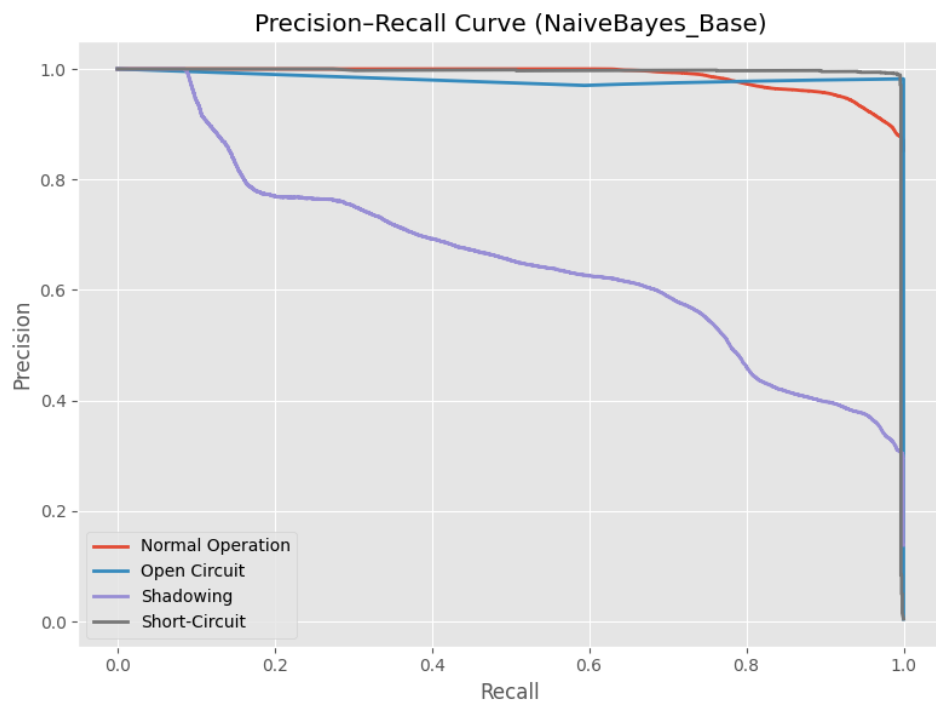


Figure 15: Naive Bayes: Precision–Recall Curve

2.5 Random Forest

Table 5: Random Forest: Evaluation Metrics

Metric	Value
accuracy	0.998
balanced accuracy	0.999
precision macro	0.998
recall macro	0.999
f1 macro	0.997
cohen kappa	0.002
log loss	1.000
roc auc macro	1.000
pr auc macro	1.000

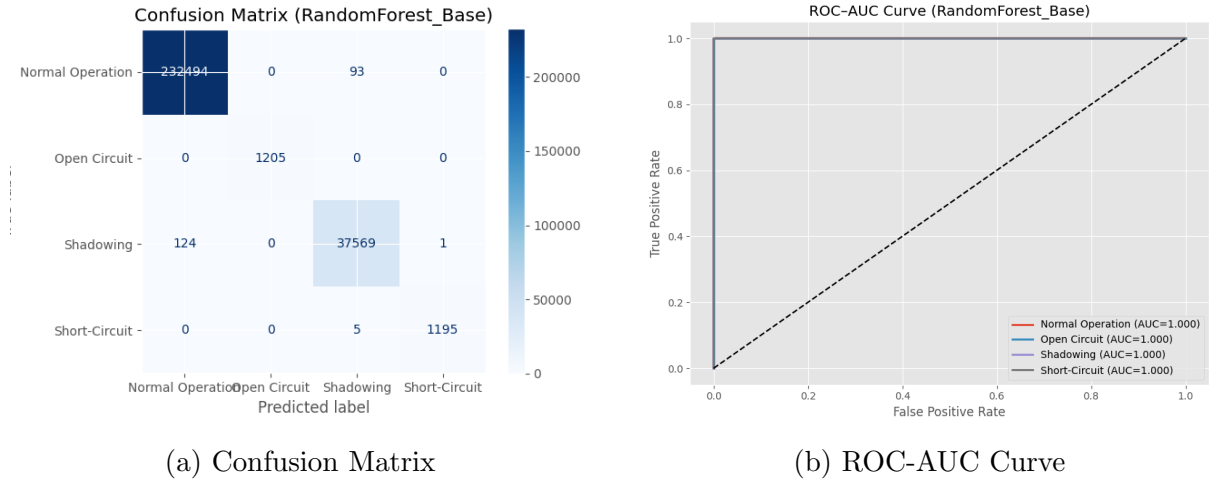


Figure 16: Random Forest: Classification Performance

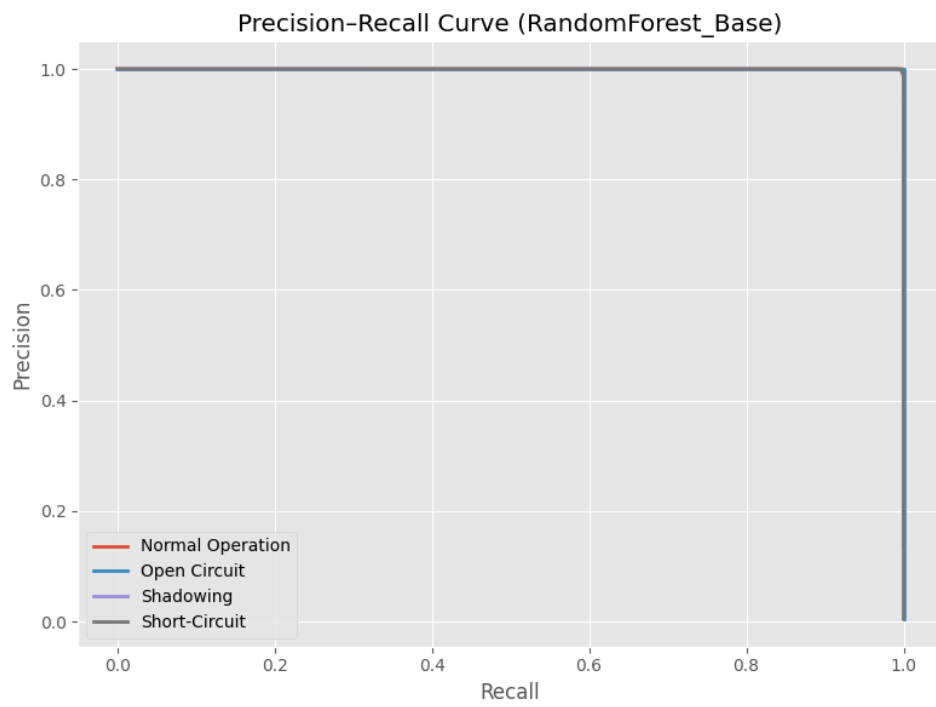


Figure 17: Random Forest: Precision–Recall Curve

2.6 Extra Trees

Table 6: Extra Trees: Evaluation Metrics

Metric	Value
accuracy	0.998
balanced accuracy	0.999
precision macro	0.998
recall macro	0.998
f1 macro	0.997
cohen kappa	0.003
log loss	1.000
roc auc macro	1.000
pr auc macro	1.000

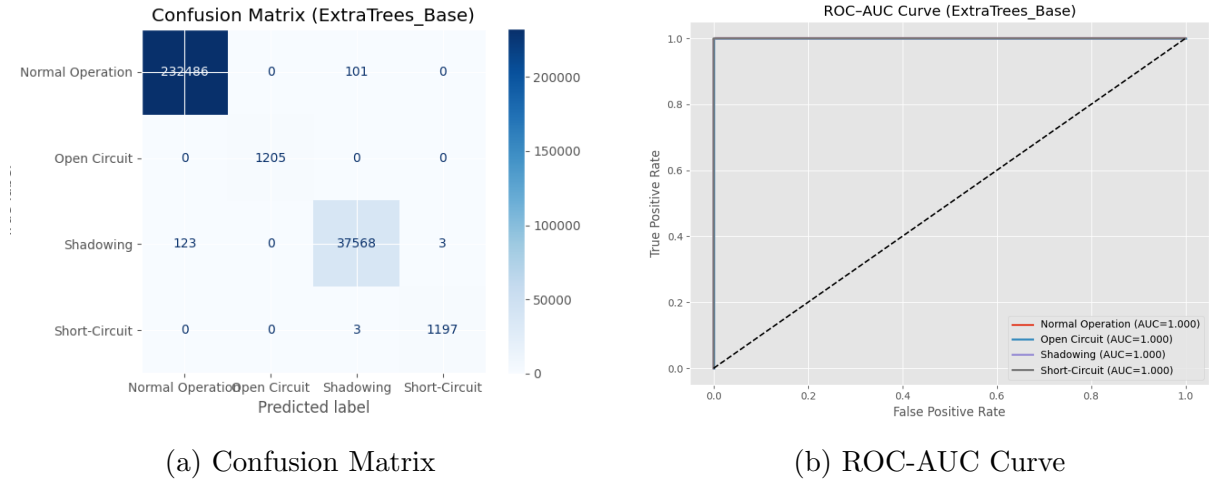


Figure 18: Extra Trees: Classification Performance

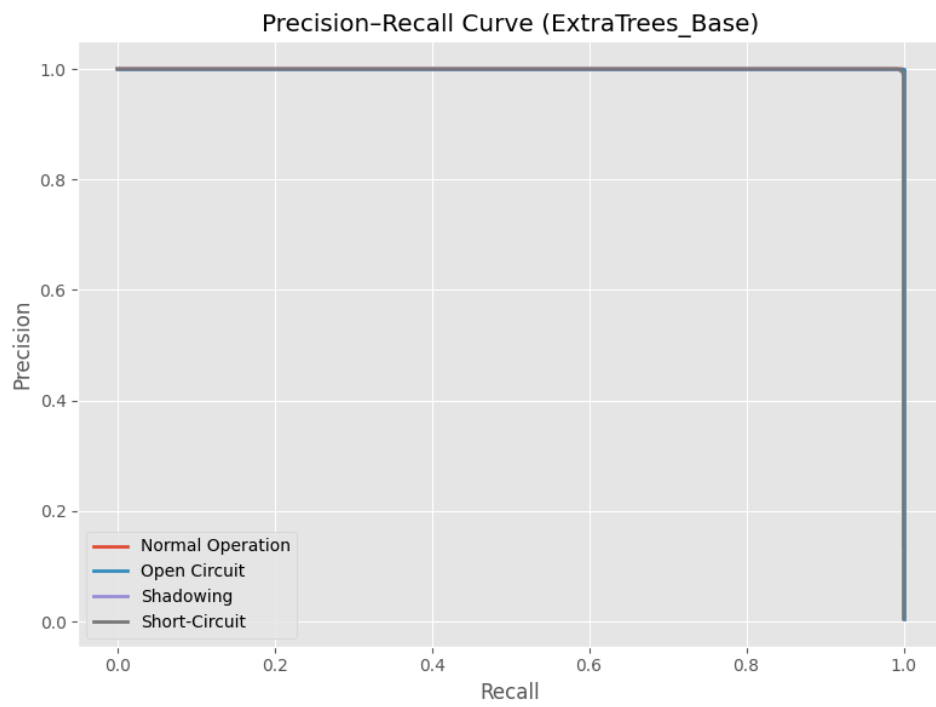


Figure 19: Extra Trees: Precision-Recall Curve

2.7 Logistic Regression

Table 7: Logistic Regression: Evaluation Metrics

Metric	Value
accuracy	0.957
balanced accuracy	0.885
precision macro	0.957
recall macro	0.910
f1 macro	0.675
cohen kappa	0.267
log loss	0.983
roc auc macro	0.952
pr auc macro	0.952

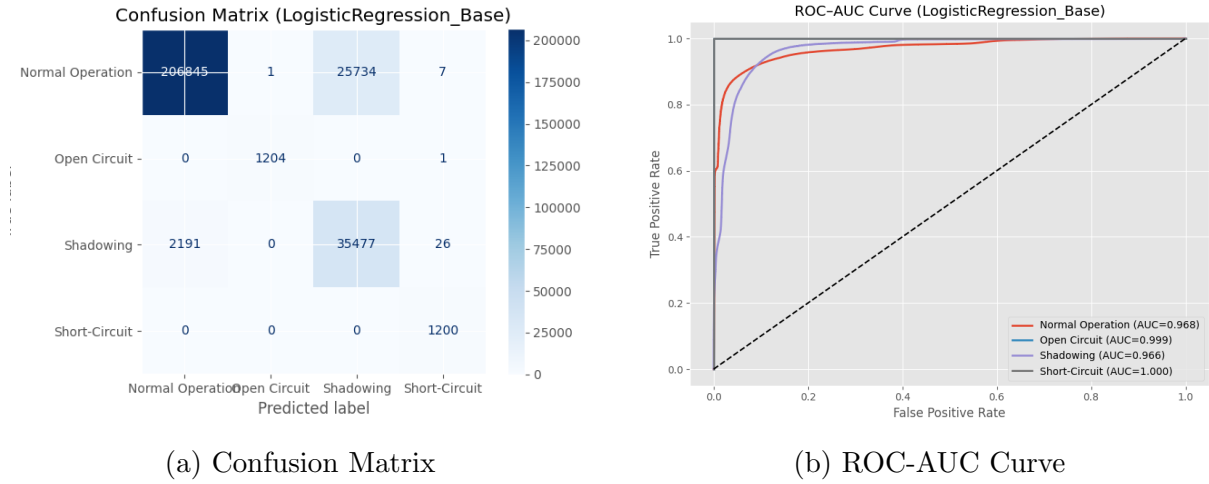


Figure 20: Logistic Regression: Classification Performance

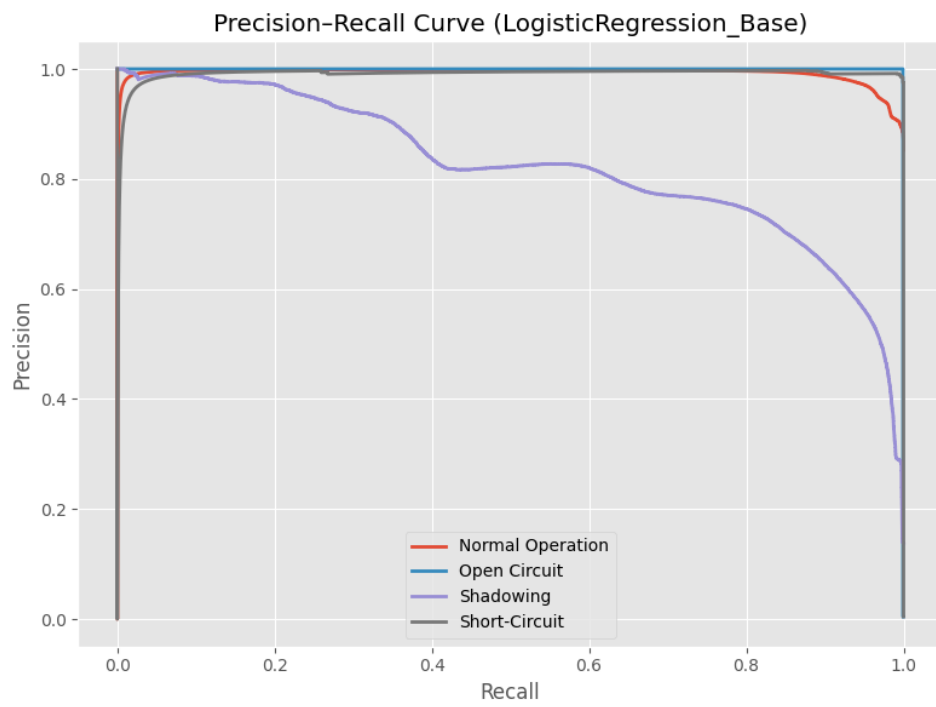


Figure 21: Logistic Regression: Precision–Recall Curve

2.8 Decision Tree

Table 8: Decision Tree: Evaluation Metrics

Metric	Value
accuracy	0.997
balanced accuracy	0.997
precision macro	0.997
recall macro	0.997
f1 macro	0.996
cohen kappa	0.040
log loss	0.998
roc auc macro	0.994
pr auc macro	0.994

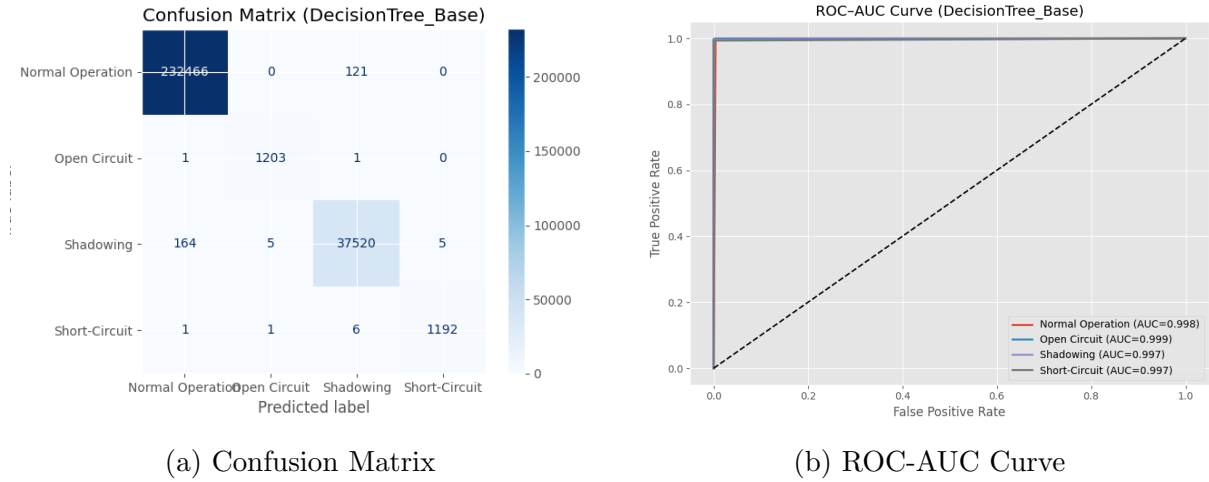


Figure 22: Decision Tree: Classification Performance

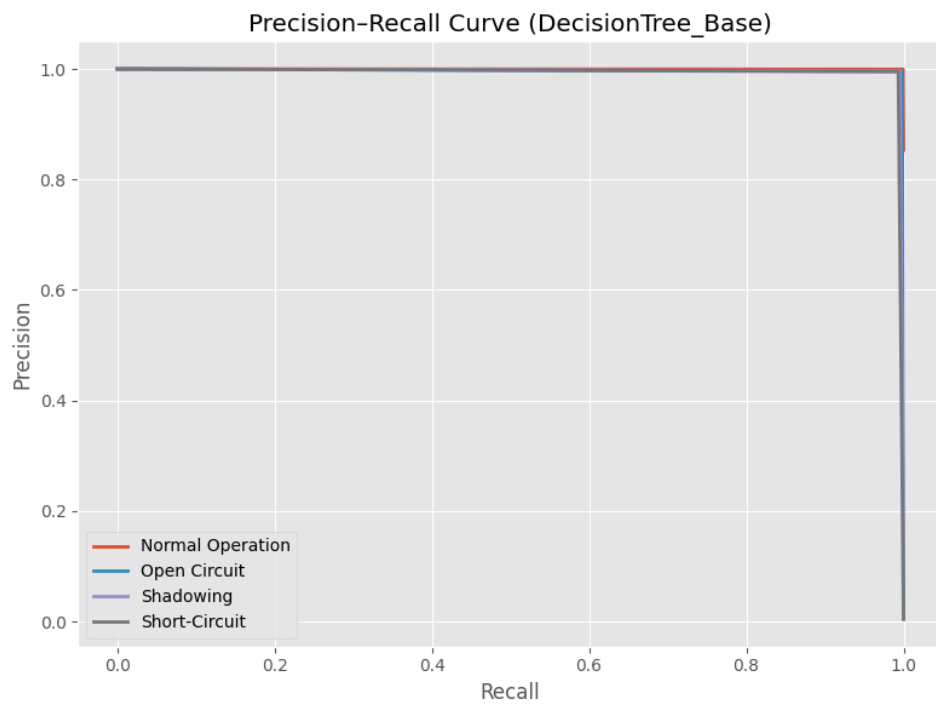


Figure 23: Decision Tree: Precision–Recall Curve